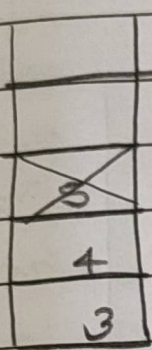


Monotonic stack ::

Initially let's know about stack, (3)

stack :: it follows the principle of LIFO (Last In First Out)



push (3)
push (4)
push (5)
pop () // 5

Monotonic stack :: It is a stack that maintains elements in increasing or decreasing order

- 2 types ::
- * Monotonic increasing stack
 - * Monotonic decreasing stack

① Monotonic Increasing stack ::

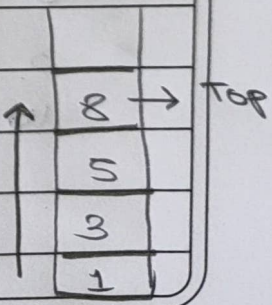
Elements inside stack are always increasing from bottom to top.

eg :: stack :: 1

3 1 < 3 < 5 < 8

5
8

Top :: 8



when to use ? * next smaller element
* previous smaller element

② Monotonic Decreasing Stack

Elements inside stack are always decreasing from bottom to top.

eg:- stack :: 8

5

3

1

$\therefore 8 > 5 > 3 > 1$

Top :: 1

1	→ Top
3	
5	
8	

when to use ? * next greater element
* previous greater element

why use a monotonic stack ?

Suppose : arr = [4, 5, 2, 10]

Find next greater element for every element ;

4 → search right side greater element = 5

5 → search right side 10

2 → search right side 10

10 → search right side but no element

But, the above searching is done using brute force method

where, for every element we scan the array.

$$N \text{ elements} * N \text{ searching} = O(N^2)$$

Therefore, for 10000 elements, it is not possible.

Monotonic stack optimization: Every element only needs to be pushed once and removed once.

Therefore; Push operation = N times

Pop operation = N times

$$\text{Total} \therefore O(2N) = O(N)$$

$$(0.1 + 1) = 1.1 \text{ correct}$$

How to identify monotonic stack problems?

next greater

// monotonic decreasing stack

next smaller

// monotonic increasing stack

distance / span

↳ eg: how many days until warmer temp?

histogram / boundaries

↳ eg: largest rectangle in histogram.

$$O(N)$$

$$O(N)$$

Problem 1: Next Greater Element I

(Leetcode 496)

2 distinct integer arrays

nums1 $\rightarrow$ a subset of nums2

nums2 $\rightarrow$ the main array

every element in nums1, find its ^{next} greater element in nums2.

the next greater element of an element x is the first element to the right of x in the same array that is greater than x .

If there is no greater element on the right side,

return -1.

eg:- nums1 = [4, 1, 2]

nums2 = [1, 3, 4, 2]

we need to find the next greater element for each element of nums1 by searching in nums2.

4 :: nums2 = [1, 3, 4, 2]

↑
[2] $\rightarrow$ no

NGE(4) = -1

For 1 :- num2 = [1, 3, 4, 2] $\uparrow$ 1 count

nge(1) = 3

For 2 :- num2 = [1, 3, 4, 2]

nge(2) = -1

Output :- [-1, 3, -1]

eg 2 :- num1 = [2, 4]

num2 = [1, 2, 3, 4]

For 2 :- num2 = [1, 2, 3, 4] $\uparrow$ 3 count

nge(2) = 3

For 4 :- num2 = [1, 2, 3, 4]

nge(4) = -1

Output :- [3, -1]

Important observation :-

① num1 element are always present in num2

② we only look towards right side

Dry run using monotonic stack ::

nums1 = [4, 1, 2] nums2 = [1, 3, 4, 2]

Start :: Stack = []

2 :: NGE(2) = -1

Stack = [2]

4 :: NGE(4) = -1

4 > 2 ∴ stack = [4]

3 :: NGE(3) = 4

3 < 4 ∴ stack = [4, 3]

1 :: NGE(1) = 3

[4, 3, 1] ∴ stack = [4, 3, 1]

Final map ::

1 → 3

3 → 4

4 → -1

2 → -1

Now for nums1 :: [4, 1, 2]

Look into map ::

4 → -1

1 → 3

2 → -1

Output :: [-1, 3, -1]

Problem Q :: Daily temperatures
(LeetCode, 739)

eg :: temperature = [30, 40, 50, 60]

30 → 1

40 → 1

50 → 1

60 → 0

output = [1, 1, 1, 0]